# NUMERICAL METHODS FOR THE DESIGN AND ANALYSIS OF WINGS AT SUPERSONIC SPEEDS

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## SUMMARY

In rather extensive employment of numerical methods for the design and analysis of arbitrary-planform wings at supersonic speeds, certain deficiencies have been revealed, particularly in application to wings with slightly subsonic leading edges. Recently devised numerical techniques which overcome the major part of these deficiencies have now been incorporated into the methods. In order to provide a self-contained description of the revised methods, the original development as well as the more recent revisions have been subjected to a thorough review in this report.

Revisions to the wing-design method have virtually eliminated irregularities that often arose in the definition of the camber surface in the immediate vicinity of the wing leading edge. An aft-element sensing technique has been incorporated in the analysis method to suppress pressure oscillations which formerly required application of a powerful nine-point smoothing formula. These improvements, in combination with more compatible summation methods in the design and analysis mode, have reduced small but disturbing discrepancies which sometimes arose between wing loadings and forces determined for an optimized wing and loadings and forces calculated for that same shape upon submittal to the evaluation program.

## INTRODUCTION

Because of its simplicity and versatility, linearized theory has been employed rather extensively in the design and analysis of supersonic aircraft. Descriptions of design and analysis methods based on linearized theory and examples of results obtained in application to typical problems are given in references 1 to 4.

One important application of linearized-theory methods is in the design of wing lifting surfaces for drag minimization and in the analysis of pressure loadings and overall forces on wings of specified shape. Although exact linearized-theory solutions are available for certain problems, for example, definition of minimum-drag surfaces for delta or arrow planforms and evaluation of loadings and forces on flat-plate wings of simple planform, these solutions are not directly applicable to the complex wing planforms